Class Bot — User Guide

NUBTK · CSE · Shovon Mandal (SM) · Fall 2026

How to run the Discord class server every day: attendance, announcements, students,
Excel files, updates and fixing problems.

1. The big picture
2. How students join
3. The control panel
4. Attendance
5. Announcements & messages
6. Students & Excel
7. Fixing a student
8. Marks by private DM
9. Changing settings & updating
10. Tinker Board care
11. All commands (with examples)
12. Problems & fixes
13. Routine & reminders

1. The big picture

The bot runs **24/7 on the Tinker Board**. You control it from Discord, in the private channel `🛠️|admin`, which only you can see. Code lives on GitHub; you edit it on your PC.

```
PC (edit)  ──git push──▶  GitHub  ──auto, within 5 min──▶  Tinker Board (bot runs here)
```

Server part	What it is for
👋 WELCOME	New students only. Section buttons to register. Hidden after they verify.
🏫 ALL STUDENTS	📌 notice-board (everyone) and 🔊 lounge.
🌐 Course groups	One per course, shared by its sections (e.g. CSE 1205/1206 · DLD · 2C, 2D).
📘 Sections (2C, 2D, 3D, 3E, 4B, 4P)	📢 announcement · 💬 general-text ✍️ · 📚 resources · 📊 marks-sheet · 🎞️ presentation-slide ✍️ · 🗓️ routine · 🧪 lab (DLD/OOP only) · 🔊 voice channel
🔒 ADMIN	🛠️ admin (panel + commands) and 📜 bot-logs (automatic logs, daily backup).

> ✍️ Students can write, attach files and create threads **only** in **general-text** and **presentation-slide**. Everywhere else they can read and react.

2. How students join

1. You click 🔗 **Invite link** in the panel and post the ready message in the Facebook / WhatsApp class group.
2. The student opens the link, goes to `👋|welcome` and clicks **their section button**.

3. A form opens: **Student ID** (11 digits) and **full name** → Submit.
4. Done. They get their section role, the nickname `1964_2C_Rahim Uddin`, a "👉 Start here" link to their section's pinned welcome card, and the welcome area disappears.

Backup way: a student can type `2C 11250121964 Rahim Uddin` in `🪪|id-name-section`. If a student leaves and comes back, their section is restored automatically.

> 🟡 If someone uses a Student ID that another account already used, they are **not** verified automatically. You get an **Approve / Reject** message in the admin channel.

3. The control panel

The panel is a set of messages in `🛠️|admin`: **each row is one message** — it lists what every button does, and **the buttons are right under that text**. Click a button → a **private menu** opens (only you see it) → choose from dropdowns or fill a short form → the result is posted in the admin channel.

> 🎛️ **Every bot message** in the admin channel ends with two small buttons: 🎛️ **Panel** (opens the whole panel privately, right there — no scrolling) and 📖 **All commands** (every text command with an example).

Row 1 · Communication

📢 Announce | ✉️ DM students | 🗓️ Scheduled posts | 🔗 Invite link | 📎 Marks / roster

Row 2 · Attendance

▶️ Start | ⏹️ Stop | ⏱️ Running now | 📈 Report | 📊 Stats

Row 3 · Students

📋 Student list | 📗 Excel | 🔎 Find | 🟡 Pending | 🔔 Remind unverified

Row 4 · Manage members

✅ Verify | 🔀 Move section | ✏️ Rename | ❌ Unverify | 🚪 Kick unverified

Row 5 · System

📌 Re-post welcome | 💾 Backup | 🖥️ Status | 🔄 Update | ♻️ Restart

Row 6 · Routine & reminders

🔄 Check routine | 🗓️ Show routine | ⏰ Remind a message

Row 7 · Help

📖 All commands | 🎛️ Show panel here

Panel missing or old? Type `!panel` in the admin channel to post it again.

4. Attendance

Take attendance in a class

1. Click ▶️ **Start attendance**.

2. Choose **section** → **course** (for lab sections choose theory or lab) → **auto-stop time** (5–120 min, or no auto-stop) → **Start**.
3. The bot posts "Attendance is open" in that section's announcement channel and pings the students.
4. Students join the section's 🔊 **voice channel**. The bot records when each student joins and leaves.
5. Click ⏹️ **Stop attendance** (or let auto-stop do it). An **Excel sheet** is posted.

What the Excel sheet shows

Top part	Course, section, date, start time, stop time, **class duration**, taken by, present count
Each student	Status (**P** / **A**), first join, last leave, **minutes in class**, % of class time, number of joins

Leaving and joining again is added correctly. Students of another section who joined are listed as `P*`.

Semester report

Click 📈 **Report** → section → course. You get an Excel file with one column per class (`P (42m)` or `A`), total present and %. Students **below 60%** are marked red.

> 💡 Want a minimum stay? In `config.json` set `"attendance_min_percent": 50`. Then a student must stay at least half the class to be counted present.

Commands: `!att_start 2C 15` · `!att_start 3D CSE2110 30` · `!att_stop 2C` · `!att_status` · `!att_report 2C CSE1205`

5. Announcements & messages

Post an announcement

1. Click 📢 **Announce**.
2. Choose one or more targets: All students, a section, a section's lab channel, or a course (all its sections).
3. Click **Write & ping** (students get a notification) or **Write (no ping)**.
4. Type a title and the message → Submit.

Post later (scheduled)

In the same form, fill the **Schedule** box: `2026-10-26 09:00`, `21:30` (today or tomorrow) or `+2h` / `+30m`. See or cancel waiting posts with 🗓️ **Scheduled posts**.

Announcement with a file (PDF, slide, image)

Buttons can't carry files, so type the command and attach the file to the same message:

```
!announce 2C Lecture 5 slides are attached
```

Private DM to students

Click ✉️ **DM students** → choose targets → write the message. Every student gets it privately (about 1 second per student).

6. Students & Excel

I want to...	Do this
See how many students	📊 **Stats**
See a section's students	📋 **Student list** → section
Get the list in Excel	📗 **Excel** → Everything / a section / a course. One sheet per course + section, plus a Summary sheet.
Find one student	🔎 **Find** → ID, name or Discord name
Remind people who didn't verify	🔔 **Remind unverified** → they get a DM and a mention
Check who hasn't joined at all	Upload the official list once: `!roster_upload 2C` + attach CSV (ID, Name). Then `!missing 2C`.

7. Fixing a student

Problem	Button	What happens
Can't verify / wrong info	✅ **Verify** → member + section → ID + name	Verified by you
Chose the wrong section	🔀 **Move section** → member + section	Roles and nickname change
Name spelled wrong	✏️ **Rename** → member → new name	Nickname fixed
Not your student	❌ **Unverify** → member	Access and record removed
Same ID used twice	🟡 **Pending** → Approve / Reject	You decide
Joined but never verified	🚪 **Kick unverified** → days → Show list → Kick	They are removed

8. Marks by private DM

Each student receives **only their own marks**. Nobody else can see them.

1. Make an Excel or CSV file. First row = column names. One column must be the Student ID.
2. In `🛠️┃admin` type the command and attach the file to the same message:

```
!marks_dm CT-1 Marks
```

Student ID	Name	CT-1 (10)	Attendance (5)	Total
11250121964	Rahim Uddin	8	5	13
11250121965	Karima Akter	9	4	13

The bot reports how many were sent and which IDs are not registered on the server.

9. Changing settings & updating

Settings are in `config.json` (on your PC, in the project folder). Common changes:

Change	Where in config.json	After pushing
Your info, email, WhatsApp	`"instructor"`	nothing (automatic)
Invite link	`"invite_link"` (empty = bot makes one)	nothing (automatic)
Channel names / icons	`"channels"`	`!setup_server`
Add / remove a section or course	`"sections"`, `"courses"`	`!setup_server`
Attendance minimum %	`"attendance_min_percent"`	nothing

How to send a change to the Tinker Board

Open the project folder → click the address bar → type `cmd` → Enter. Then:

```
git add .
git commit -m "what I changed"
git push
```

Within **5 minutes** the bot pulls the change and restarts by itself (a message appears in 📜 bot-logs). In a hurry? Click 🔄 **Update**. If the new code or config has a mistake, the bot **does not** restart and tells you the error.

> 🛑 **Never** type `!reset_server` unless you really want to delete **every** channel and message. Both `!setup_server` and `!reset_server` ask **Yes / Cancel** first — read the warning.

New semester

1. Edit `semester`, `courses` and `sections` in config.json.
2. git push → wait for the update → type `!setup_server` → Yes.
3. Click 🔗 **Invite link** and share it with the new students.

10. Tinker Board care

From your PC's cmd: `ssh shovon@192.168.0.221`, then:

Command	What it does
`bash ~/discord-bot/scripts/bot.sh status`	Is the bot running?
`bash ~/discord-bot/scripts/bot.sh log`	Live log (Ctrl+C to exit)
`bash ~/discord-bot/scripts/bot.sh restart`	Restart the bot

Command	What it does
`bash ~/discord-bot/scripts/bot.sh update`	Pull from GitHub + restart (if !update fails)
`sudo systemctl stop nubtk-bot`	Stop the bot

⚠️ **Run the bot in one place only.** If it runs on the PC and the Tinker Board at the same time, everything happens twice and the data splits.
To move it to the PC: stop it on the board, then copy the data: `scp -r shovon@192.168.0.221:~/discord-bot/data .`

💾 A backup zip of all data is posted in 📜 bot-logs every night at 23:30. Click 💾 **Backup** any time for a fresh one. Student data never goes to GitHub.

11. All commands (with examples)

Type these in `🛠️┃admin`. `!help` shows the same list in Discord.

Command	What it does	Example
🏗️ Setup		
`!panel`	Re-post the button control panel	`!panel`
`!setup_server`	Create / sync all channels & roles (asks to confirm)	`!setup_server`
`!setup_welcome`	Re-post welcome buttons + instructor cards	`!setup_welcome`
`!invite`	Invite link + join guide for students	`!invite`
`!reset_server`	⚠️ Delete EVERYTHING, build fresh (asks to confirm)	`!reset_server`
🎓 Students		
`!stats`	Count per section & course	`!stats`
`!list <sec>`	All students of a section	`!list 2C`
`!find <id/name>`	Search a student	`!find 11250121964 · !find Rahim`
`!verify @user <sec> <id> <name>`	Verify manually	`!verify @rahim 2C 11250121964 Rahim Uddin`
`!unverify @user`	Remove access + record	`!unverify @rahim`
`!move @user <sec>`	Change section	`!move @rahim 2D`
`!rename @user <name>`	Fix name / nickname	`!rename @rahim Md Rahim Uddin`
`!pending`	Requests waiting for approval	`!pending`
`!notify_unverified`	Remind unverified members	`!notify_unverified`
`!kick_unverified <days> [confirm]`	List / kick never-verified	`!kick_unverified 7` → `!kick_unverified 7 confirm`
`!roster_upload <sec>`	Upload official list (attach CSV)	`!roster_upload 2C` + attach `2C.csv`
`!missing <sec>`	Roster students not joined yet	`!missing 2C`
📢 Communication		
`!announce <target> <text>`	Post + ping (all / 2C / 2C:lab / CSE1205)	`!announce 2C Class test on Sunday at 10 AM`
`!announce_silent <target> <text>`	Same, no ping	`!announce_silent all Slides are uploaded`

Command	What it does	Example
`!dm <target> <text>`	DM every student	`!dm 4P Tomorrow's class is online`
`!marks_dm <title>`	Send each student their own marks (attach CSV/Excel)	`!marks_dm CT-1 Marks + attach ct1.xlsx`
`!scheduled`	List / cancel scheduled announcements	`!scheduled`
🗓️ Routine & reminders		
`!routine_check`	Read the routine sheet, update pinned routines, announce changes	`!routine_check`
`!routine [sec]`	Show the saved routine	`!routine 2C`
`!routine_repost`	Re-post every section's pinned routine	`!routine_repost`
(right-click a message → Apps → ⏰ Set reminder)	Re-announce a message now or later	`deadline / CT reminder`
🗓️ Attendance		
`!att_start <sec> [course] [minutes]`	Start voice attendance	`!att_start 2C 15` · `!att_start 3D CSE2110`
`!att_stop <sec>`	Stop + get CSV	`!att_stop 2C`
`!att_status`	Running sessions	`!att_status`
`!att_report <sec> [course]`	Semester % per student	`!att_report 2C CSE1205`
💾 Data & System		
`!excel [target]`	Student list as Excel	`!excel` · `!excel 2C` · `!excel CSE2217`
`!export [target]`	Student CSV/JSON files	`!export CSE1205 2C`
`!backup`	Zip of all data	`!backup`
`!status`	Uptime, CPU temp, RAM	`!status`
`!logs [lines]`	Last log lines	`!logs 50`
`!update`	Pull from GitHub + restart	`!update`
`!restart`	Restart the bot	`!restart`

12. Problems & fixes

Problem	Fix
Bot is offline	Check the Tinker Board has power and internet. SSH in → `bot.sh status` → `bot.sh restart`.
Buttons say "interaction failed"	The bot was restarting. Wait 20 seconds and try again. Still failing → `!panel`.
Commands get no reply	Type them in `🛠️|admin`, not in another channel.
Channels appear twice	The board did not have the `data` folder. Run `!reset_server` once → Yes.
Student says "I can't see my channels"	🔎 **Find** them. Not verified → ✅ **Verify**. Wrong section → 🔀 **Move**.
Student didn't get a DM	Their Discord privacy blocks DMs from servers. Ask them to allow it (Server → Privacy Settings → Direct Messages).
Update didn't arrive	Check `git push` worked. Click 🔄 **Update** and read the message.
Bot can't give roles / change nicknames	Server Settings → Roles → drag the bot's role to the top.

13. Routine & reminders

Class routine (from the department Google Sheet)

1. Click 🔄 **Check routine** (it also runs by itself every 6 hours).
2. The bot reads the official routine sheet and finds **your** classes (SM) for every section.
3. Each section's `🗓️|routine` channel keeps **one pinned message = the latest routine** (day, time, course, room).
4. If something changed, the old pinned message is replaced and the bot posts what changed: ➕ new · ➖ removed, and pings the section.

> ⚠️ The Google Sheet must be shared as **Anyone with the link → Viewer**, otherwise the bot cannot read it. The link is in `config.json → routine.sheet_url`.

Reminder for a message (deadline, CT...)

1. **Easiest:** right-click the message (e.g. the assignment announcement) → **Apps** → ⏰ **Set reminder**.
   Or click ⏰ **Remind a message** in the panel and paste the message link (right-click → Copy Message Link).
2. **When?** leave empty to post now, or write times: `2026-10-28 20:00, 2026-10-29 08:00` · `+1d` · `21:30`.
3. Optional note, e.g. "Assignment 2 deadline is tomorrow 11:59 PM!".

4 **Where?** empty = the same section or course as the message, or type `2C`, `CSE1205`, `all`.

The reminder is posted in the announcement channel with a ping, a quote of the message and an **Open the original message** button. Waiting reminders are listed under 🗓️ **Scheduled posts**, where you can cancel them.

No "Apps → ⏰ Set reminder" in the right-click menu? Open this link once and click Authorize (the bot stays in the server):
`discord.com/oauth2/authorize?client_id=1297622424640294922&permissions=8&scope=bot+applications.commands`

Code: github.com/shovon-mandal/academic-discord-management-bot · Secret token is only in the `.env` file — never share it or upload it.